

Midterm 1 Reflection

Note the questions you got wrong or lost some points on and consider why that's the case. Here is a list of some potential reasons:

- a) Misunderstood the question
- b) Didn't know the material/unfamiliar with the concept
- c) Difficulty applying a concept to a new context
- d) Overthought the question/thought it was a trick question
- e) Didn't know where to start
- f) Careless mistake
- g) Running out of time and had to rush
- h) Other (explain)

Question #	Why I didn't get it completely right	Main concept that question is testing	Notes on how to correct solution
1	I overcomplicated the statement by adding extra variables and conditions that weren't part of the question. I also introduced y_1 and y_2 , which weren't defined, making the logical structure unclear.	Predicate logic and quantifiers — specifically, expressing English statements formally using universal and existential quantifiers.	Stick closely to the meaning of the sentence and translate it directly into logic without adding unnecessary complexity. With the simpler form being in the exam solutions PDF, I should focus on the structure and scope of quantifiers rather than trying to encode extra assumptions about any distinctness.
2	I only proved one direction of the equality instead of both. I showed $A \cap B \subseteq A$ but didn't show $A \subseteq A \cap B$.	Set theory and methods of proof — specifically, proving set equality by demonstrating both subset relations.	When proving two sets are equal, always structure the argument as two separate parts: $A \cap B \subseteq A$ $A \subseteq A \cap B$ Each direction must be clearly justified with the definition of subset and intersection. Writing "if and only if" statements can also help keep the logic consistent.
3	I forgot to include the summation notation in the inductive hypothesis and my algebra in the inductive step was incorrect. I also didn't carefully expand and simplify the new term when going from k to $k + 1$.	Mathematical induction — specifically how to correctly apply the inductive hypothesis to summations and simplify expressions to match the formula for $n = k + 1$.	Write the statement clearly in summation form before starting the proof. In the inductive step, substitute the hypothesis $\sum_{m=1}^k \frac{1}{(3m-2)(3m+1)} = \frac{k}{3k+1}$ and carefully add the next term $\frac{1}{(3(k+1)-2)(3(k+1)+1)}$. Simplify symbolically until it matches $\frac{k+1}{3(k+1)+1}$. Keep algebra neat and structured — skipping steps caused mistakes.

4	I started using generating functions instead of solving the recurrence directly with the characteristic equation. My setup became messy and incomplete because I didn't fully isolate or solve for constants, and I ran out of time before finishing the algebra.	Solving non-homogeneous linear recurrence relations using the characteristic equation and finding both the homogeneous and particular solutions.	Use the standard approach: Solve the homogeneous part $a_n - 4a_{n-1} + 4a_{n-2} = 0$. Find a particular solution for the 3^n term by substitution. Combine them and use initial conditions to solve for constants. Avoid switching methods mid-solution unless the question asks for it — generating functions are valid but slower under time pressure.
5	I added the generating functions instead of multiplying them. I wrote $\frac{1}{1-x} + \frac{1-x^2}{1-x} + \frac{1}{1-x^2}$ when I should have used the product to encode choices for Alice, Ben, and Chloe simultaneously. I also didn't extract the coefficient of x^n at the end.	Generating functions for constrained distributions (stars-and-bars with caps/parity). Product principle and coefficient extraction.	Use a product, one factor per child: $A(x) = \underbrace{\frac{1}{1-x}}_{\text{Alice}} \cdot \underbrace{(1+x+x^2)}_{\text{Ben}} \cdot \underbrace{\frac{1}{1-x^2}}_{\text{Chloe}}$ $= \frac{1+x+x^2}{(1-x)^2(1+x)}.$ Do partial fractions: $\frac{1+x+x^2}{(1-x)^2(1+x)} = -\frac{3}{4} \frac{1}{1-x} + \frac{3}{2} \frac{1}{(1-x)^2} + \frac{1}{4} \frac{1}{1+x}.$ Extract the coefficient of x^n using what is shown in the solutions PDF to get: $A_n = \frac{3}{2}n + \frac{3+(-1)^n}{4}$ So the number of distributions is a_n above.
6	I wrote a recurrence and jumped to the answer without justifying why the recurrence matches the algorithm or which Master Theorem case applies. The answer wanted the rationale that connects the bitonic-peak search to halving	Setting up recurrences from an algorithm description, then applying the Master Theorem correctly (identify a , b , d and the case).	Each step compares $A[\lfloor n/2 \rfloor]$ and $A[\lfloor n/2 \rfloor + 1]$ in $O(1)$ time. Based on the comparison, we recurse on exactly one half (left if decreasing at mid, right if increasing). Base case when the range size is 1: $T(1) = \Theta(1)$.

	work each step and the constant work per step.		So the recurrence is: $T(n) = T(2n) + \Theta(1), T(1) = \Theta(1).$ Apply Master Theorem with $a = 1$, $b = 2$, $d = 0$. Since $a = b^d = 1$, it's Case 2: $T(n) = \Theta(n \log n) = \Theta(\log n).$ Add a one-sentence interpretation: the problem size halves each time, so the number of levels in the recursion tree is $\lfloor \log_2 n \rfloor$, and each level costs constant time, giving $O(\log n)$.
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Reflect on your test preparation techniques to answer the following questions.

1. When did you start studying?

A week and a half before the exam is when I started studying, first three days I skimmed over notes, textbook, and homeworks to review before starting the practice exam. Afterwards I started doing the practice exam daily as well as continuing to look at notes and homeworks afterwards the week prior to the exam.

2. How did you study?

I studied by skimming through textbook and homeworks. Doing the practice exam a few times and going through the notes moderately. I attempted to prioritize topics on the practice exam and some topics on the homeworks that I didn't understand in the notes.

3. What techniques from your studying were most helpful?

The practice exam was really good at helping out with the questions that were on the actual exam. So I attempted the practice exam a couple times (5 times too many?). Skimming through the textbook didn't do much but I probably should have went through it more.

4. What was least helpful?

The homework's were good for practice but not much of the problems on there helped with solving the problems on the exam. That is a given since we were tested on how well we could work through and think about a problem conceptually. However I looked over homework's primarily to memorize how to do the problems and that was a bad idea.